1 $[ML^{-1}T^{-1}]$ is the dimensional formula of _____

A. Surface Tension
B. Angular Momentum
C. Stress
D. Coefficient of Viscosity

2 Most accepted theory of the origin of the universe is _____

A. Big Bang theory
B. Steady State theory
C. String Theory
D. Static Universe Theory

3 P-type semiconductor is made by doping trivalent impurity with germanium. Which of the following doping makes P-type semiconductor?

A. Arsenic
B. Phosphorous
C. Indium
D. Antimony

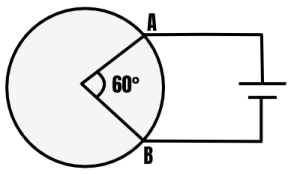
4 Unit of self inductance is _____

A. $\frac{\text{Volt} \cdot \text{amp}}{\text{sec}}$
B. $\frac{\text{Volt} \cdot \text{sec}}{\text{amp}}$
C. $\frac{\text{Volt}}{\text{amp}}$
D. $\frac{\text{Joule}}{\text{amp}}$

5 Choose the wrong.

A. Luminous flux = lumen
B. Luminous intensity = candela
C. Intensity of illumination = lux
D. Luminous flux = $\frac{\text{lumen}}{\text{m}^2}$

6 A uniform wire of resistance 36 Ω is bent in the form of a circle. A battery is connected between positions A and B as shown. The effective resistance between A and B is _____



A. 18 Ω
B. 6 Ω
C. 30 Ω
D. 5 Ω

7 Resolving power of telescope is _____

A. $\frac{1.22}{\lambda D}$
B. $\frac{1.22\lambda}{D}$
C. $\frac{D}{1.22\lambda}$
D. $\frac{D\lambda}{1.22}$

8 The work function of sodium is 2eV. When the light of wavelength 500 nm is incident on its surface, the maximum velocity of photoelectrons will be _____

A. 413 $\frac{\text{km}}{\text{s}}$
B. 430 $\frac{\text{km}}{\text{s}}$
C. 513 $\frac{\text{km}}{\text{s}}$
D. 220 $\frac{\text{km}}{\text{s}}$

9 Find the wavelength of an electron moving with a velocity of 300 $\frac{\text{km}}{\text{s}}$.

A. 2.9093×10^{-12} m
B. 2.427×10^{-9} m
C. 3.55×10^{-11} m
D. 2.7×10^{-13} m

10 Half life = 1.28×10^{-5} seconds, the average life is _____

A. 1.84×10^{-5} seconds
B. 1.58×10^{-5} seconds
C. 1.28×10^{-5} seconds
D. 1.4×10^{-5} seconds

11 Property of retentivity is used in _____

A. Transformer
B. Electromagnet
C. Permanent magnet
D. Galvanometer

12 Which one is correct regarding Zener diode?

A. Ordinary diode has high reverse breakdown voltage than Zener diode
B. Ordinary diode has low reverse breakdown voltage than Zener diode
C. Ordinary diode and Zener diode have same reverse breakdown voltage
D. Ordinary diode can be used as a Zener diode

13 Phase difference between current and voltage in LCR circuit _____

A. 0
B. $\frac{\pi}{2}$
C. 0 to $\pm\frac{\pi}{2}$
D. π

14 Minimum capacitance obtained by joining 5 capacitors of 1 μF each is _____

A. 0.2 μF
B. 1 μF
C. 0.5 μF
D. 1.5 μF

15 Which is correct about achromatic combination of lens?

A. $\frac{f_1}{f_2} = \frac{\omega_1}{\omega_2}$
B. $\frac{f_1}{f_2} = -\frac{\omega_1}{\omega_2}$
C. $\frac{f_1}{f_2} = \frac{\omega_2}{\omega_1}$
D. $\frac{f_1}{f_2} = -\frac{\omega_2}{\omega_1}$

16 **Forbidden energy gap _____**

A. Maximum for conductor
B. Of insulator lies between that of conductor and semiconductor
C. Maximum for insulator
D. Minimum for semiconductor

17 **Escape velocity of earth is $11.2\ \frac{km}{s}$. Escape velocity of a planet whose g is twice that of earth and diameter is twice that of earth is _____**

A. $11.2\ \frac{km}{s}$ B. $22.4\ \frac{km}{s}$
C. $44.8\ \frac{km}{s}$ D. $33.6\ \frac{km}{s}$

18 **Resistance of two bodies r and 2r respectively, are connected in series across a battery. Find the ratio of heat consumed in 2r and r.**

A. $\frac{1}{2}$ B. $\frac{2}{1}$
C. $\frac{1}{4}$ D. $\frac{4}{1}$

19 **A person goes to office with $40\ \frac{km}{hr}$ and returns home with $60\ \frac{km}{hr}$. What is his average speed?**

A. $100\ \frac{km}{hr}$ B. $50\ \frac{km}{hr}$
C. $48\ \frac{km}{hr}$ D. $0\ \frac{km}{hr}$

20 **Penetrating power is maximum for _____**

A. Alpha ray B. Beta ray
C. Gamma ray D. X-ray

21 **A body is projected making angle 60° with vertical. What is the value of KE at the highest point?**

A. 0 B. $\frac{E}{2}$
C. $\frac{E}{4}$ D. $\frac{3E}{4}$

22 **Quark combination of proton is _____**

A. uud B. udd
C. uds D. usd

23 **A car can be stopped over a distance x when its momentum is p. What will be the stopping distance when momentum is 2p?**

A. 2x B. 4x
C. 6x D. 8x

24 **Which of the following gates can be used in the construction of digital circuit?**

A. AND B. OR
C. NAND D. NOT

25 **n capacitors with capacitance C, each connected in parallel combination across voltage V. Calculate the energy stored in the combination.**

A. $\frac{1}{2}nCV^2$ B. $\frac{1}{2n}CV^2$
C. CV^2 D. CV^4

26 **During gamma emission _____**

A. Atomic number changes by 2, mass number by 4
B. Atomic number changes by 1, no change in mass number
C. No change in atomic number and mass number
D. Atomic number and mass number both changes by 2

27 **If the length of closed and open organ pipe is same, then the ratio of fundamental frequencies is _____**

A. $\frac{1}{2}$ B. $\frac{2}{1}$
C. $\frac{3}{4}$ D. None

28 **Decreasing order of work done during an expansion process _____**

A. $W_{isobaric} > W_{isothermal} > W_{adiabatic} > W_{isochoric}$
B. $W_{isochoric} > W_{adiabatic} > W_{isothermal} > W_{isobaric}$
C. $W_{isobaric} > W_{isochoric} > W_{adiabatic} > W_{isothermal}$
D. $W_{isobaric} > W_{isochoric} > W_{isothermal} > W_{adiabatic}$

29 **Which of the following sets of temperature of source and sink has maximum efficiency?**

A. 100 K – 10 K B. 300 K – 250 K
C. 600 K – 300 K D. 300 K – 200 K

30 **A car is moving toward a cliff and blows a horn of frequency 'f' which is reflected back with frequency '2f'. Then the velocity of car is _____**

A. v B. $\frac{v}{3}$
C. $\frac{v}{4}$ D. $\frac{v}{6}$

31 **If volume of gas is suddenly reduced when temperature doubles, then ratio of final volume to initial volume will be _____**

A. $\frac{1}{2}$ B. Less than $\frac{1}{2}$
C. More than $\frac{1}{2}$ D. Between 1 and 2

32 **The cubical expansivity of the sphere is less than that of liquid. With increase of temperature, the apparent weight of sphere will _____**

A. Increase
B. Decrease
C. Remains same
D. First increase then decrease

33 A glass slab of thickness 8 cm with refractive index of 1.6 consists a layer of liquid of thickness 4.5 cm and refractive index 1.5 above the slab. What is the apparent shift of a black mark at the bottom of the slab seen from above?

A. 8 cm
B. 3.5 cm
C. 4.5 cm
D. 6 cm

34 If the temperature increases by 1°C, then the change in velocity of sound is (v at NTP = $332\ \frac{m}{s}$) _____

A. 0.4
B. 0.6
C. 0.2
D. 0.8

35 Apparent weight of a person is more than real weight when lift goes _____

A. Down with acceleration
B. Down with uniform velocity
C. Up with acceleration
D. Up with uniform velocity

36 Choose the incorrect statement.

A. Coherent source is the only required criteria for interference of light
B. Newtons ring is an example of interference
C. Diffraction takes place from coherent source from a single wave
D. Saccharimeter is based on phenomenon of polarization

37 A wire having magnetic moment 'M' & length 'L' is bent in the shape of π with all three parts of same length. The net magnetic moment of the wire becomes _____

A. $\frac{M}{3}$
B. 3M
C. M
D. $\frac{M}{4}$

38 Nuclear reactor is based on _____

A. Fission
B. Fusion
C. Radioactivity
D. None

39 When a potential difference 'V' is applied across a conductor at a temperature 'T', the drift velocity of electrons is proportional to _____

A. $\sqrt{V}$
B. V
C. $\sqrt{T}$
D. T

40 Lyman series lies in _____

A. Visible
B. Infra red
C. Far IR
D. Ultraviolet

41 In nuclear reactor, graphite is used _____

A. To decrease speed of neutron
B. As a lubricant
C. Superconductor
D. Control rod

42 Which of the following is a vector quantity?

A. Electric field
B. Electric potential
C. Electric current
D. Electric charge

43 Physical quantity responsible for angular acceleration is _____

A. Force
B. Torque
C. Angular momentum
D. Moment of inertia

44 The ratio of mean value over one half cycle to the rms of a.c. of complete cycle is _____

A. $\frac{\sqrt{2}}{\pi}$
B. $\frac{2\sqrt{2}}{\pi}$
C. $\frac{\sqrt{2}}{1}$
D. $\frac{\pi}{\sqrt{2}}$

45 The universe is expanding at the rate of $70\ \frac{km/s}{Mpc}$ and a galaxy is receding with velocity $1330\ \frac{km}{s}$. Calculate the distance of galaxy from the Earth.

A. 18 Mpc
B. 19 Mpc
C. 93100 Mpc
D. 1330 Mpc

46 Magnetic field inside the solenoid is _____

A. Directly proportional to the length of solenoid
B. Directly proportional to the current
C. Inversely proportional to the number of turns on the coil
D. Directly proportional to the square of the current

47 If two bodies of masses 4M & 16M move with same kinetic energy, what is the ratio of their momentum?

A. $\frac{2}{1}$
B. $\frac{1}{2}$
C. $\frac{4}{1}$
D. $\frac{1}{4}$

48 Electric potential in an electric dipole is proportional to _____

A. $\frac{1}{r}$
B. $\frac{1}{r^2}$
C. $\frac{1}{r^3}$
D. $\frac{1}{r^4}$

49 If N water droplets combine to form a big drop, the potential energy of the big drop is _____

A. $N^{\frac{5}{3}}U$
B. $N^{\frac{1}{3}}U$
C. $N^{\frac{2}{3}}U$
D. N^2U

50 A vessel contains oil (density $0.8\ \frac{g}{cm^3}$) over mercury (density $13.6\ \frac{g}{cm^3}$). A homogeneous sphere floats with half its volume immersed in mercury and the other half in oil. The density of the material of the sphere in $\frac{g}{cm^3}$ is _____

A. 3.3
B. 6.4
C. 7.2
D. 12.8

51 **Breathalyzer is used to detect alcohol in drivers. The compound used is in it is _____**

A. HIO_3
B. Conc. H_2SO_4
C. Conc. HNO_3
D. $K_2Cr_2O_7$

52 **IUPAC name of $CH_3 - C(CH_3)_2 - CH_2 - CH_3$**

A. 2-Methylpentane
B. 1,1,1-Trimethylpropane
C. 3,3-Dimethylbutane
D. 2,2 Dimethyl butane

53 **IUPAC name of glycerol**

A. Propan-1,2,3-trialcohol
B. Propan-1,2,3-triol
C. Butan-1,2-diol
D. Propan-1,2-diol

54 **Iodoform is used as _____**

A. Antiseptic
B. Analgesic
C. Antipyretic
D. Anaesthetic

55 **Reaction of chloroform and conc. HNO_3 gives _____**

A. Nitrochloropicrin
B. Chloretone
C. Chloropicrin
D. Carbon tetrachloride

56 **Bakelite is formed by the condensation of two compounds. Which among those reacts with $FeCl_3$ to give positive test?**

A. Phenol
B. Formaldehyde
C. Benzaldehyde
D. Benzoic acid

57 **Aniline and benzene diazonium chloride reacts to form _____**

A. p-hydroxy azobenzene
B. p-amino azobenzene
C. m-hydroxy azobenzene
D. o-hydroxy azobenzene

58 **Reactivity of halide toward S_N1 reaction follows the order _____**

A. $3^\circ > 2^\circ > 1^\circ$
B. $1^\circ > 2^\circ > 3^\circ$
C. $1^\circ > 3^\circ > 2^\circ$
D. $3^\circ > 1^\circ > 2^\circ$

59 **The stereochemistry of product in S_N2 reaction is _____**

A. Walden inversion
B. Racemisation
C. Net inversion
D. Partial racemisation

60 **Which of the following gives CH_3CHO and CH_3COCH_3 on ozonolysis?**

A. 2-methyl but-2-ene
B. 1-methyl but-2-ene
C. 2-methyl but-1-ene
D. But-2-ene

61 **What is the name of a given reaction?**

$C_6H_6 + CH_3Cl \xrightarrow{\text{Anh. } AlCl_3} C_6H_5CH_3 + HCl$

A. Friedel-Craft's alkylation reaction
B. Riemer-Tiemann reaction
C. Wurtz reaction
D. Fittig reaction

62 **In a mixture of n-heptane and iso-heptane, if n-heptane is 20%, then what is the octane number?**

A. 80
B. 100
C. 20
D. 120

63 **Reaction of carboxylic acid and alcohol gives _____**

A. Ester
B. Alcohol
C. Aldehyde
D. Amide

64 **NO_2 in nitrobenzene is _____**

A. o & p directing
B. Meta-directing
C. Neutral
D. p-directing

65 **Phenol + Bromine gives _____**

A. p-bromophenol
B. o-bromophenol
C. m-bromophenol
D. Tribromo phenol

66 **Isocyanide test is for _____**

A. Primary amine
B. Secondary amine
C. Primary alcohol
D. Secondary alcohol

67 **Forward reaction is favored by increasing pressure in _____**

A. $H_2\,(g) + I_2\,(g) \rightleftharpoons 2HI\,(g)$
B. $PCl_5\,(g) \rightleftharpoons PCl_3\,(g) + Cl_2\,(g)$
C. $N_2\,(g) + O_2\,(g) \rightleftharpoons 2NO\,(g)$
D. $2SO_2\,(g) + O_2\,(g) \rightleftharpoons 2SO_3\,(g)$

68 **Turnbull's blue is _____**

A. Ferrifero-cyanide
B. Ferricyanide
C. Ferro Ferricyanide
D. Ferrous ferrocyanide

69 **Na is manufactured commercially by electrolysis of its fused chloride, the process is called _____**

A. Castner
B. Kelner
C. Solvay
D. Down

70 **Brown ring is due to _____**

A. $FeSO_4 \cdot NO_2$
B. $FeSO_4 \cdot NO$
C. $FeSO_4 \cdot N_2O$
D. $FeSO_4$

71 **Reagents used in precipitation of group III A metals _____**

A. $NH_4Cl + NH_4OH$
B. $NH_4Cl + NH_4OH + H_2S$
C. $HCl + H_2S$
D. None

72 **Favourable condition for formation of conc. H_2SO_4 by contact process is _____**

A. Low temperature, high pressure, high concentration of reactant
B. High temperature, high pressure, high concentration of reactant
C. High temperature, low pressure, high concentration of reactant
D. Low temperature, low pressure, high concentration of reactant

73 **Which of the following is a neutral oxide?**

A. CO
B. SnO
C. Al_2O_3
D. SiO_2

74 **Difference between mass number and atomic number gives _____**

A. Proton
B. Nucleus
C. Neutron
D. Positron

75 **Number of electron donor atoms in EDTA is _____**

A. 2 N and 4 O
B. 4 N and 2 O
C. 2 N and 2 O
D. 1 N and 1 O

76 **Exception to octet rule _____**

A. CO_2
B. NO_2
C. HF
D. CCl_4

77 **Galvanization is the _____**

A. Deposition of zinc on iron
B. Deposition of tin on iron
C. Deposition of copper on iron
D. Deposition of aluminium on iron

78 **Molarity of pure water is _____**

A. 0
B. 55.55
C. 45.55
D. 1

79 **Number of ionic, covalent and co-ordinate bonds present in NH_4Cl _____**

A. 1, 3, 1
B. 1, 3, 2
C. 1, 4, 0
D. 1, 3, 3

80 **The rate of diffusion of a gas at constant temperature is _____**

A. directly proportional to molecular mass
B. inversely proportional to molecular mass
C. directly proportional to square root of molecular mass
D. inversely proportional to square root of molecular mass

81 **Unit of rate constant for third order reaction is _____**

A. $\sec^{-1}$
B. $\frac{L^2}{mol^2 \cdot sec}$
C. $\frac{mol}{L \cdot sec}$
D. $\frac{mol^2}{L^2 \cdot s}$

82 **Oxidation number of carbon in CH_3Cl, CH_2Cl_2, $CHCl_3$ and CCl_4 _____**

A. -2, 0, +2, +4
B. +4, +2, 0, -2
C. -2, 2, 0, 4
D. 0, 2, 4, -2

83 **Three unpaired electrons in nitrogen is explained by _____**

A. Aufbau principle
B. Hund's rule
C. Pauli's exclusion principle
D. Le-Chatelier principle

84 **Activation energy is decreased by _____**

A. increasing concentration of reactants
B. increasing temperature
C. use of catalyst
D. decreasing concentration of reactants

85 **100 mL of 0.2 M H_2SO_4 is treated with 100 mL of 0.2 M NaOH. Calculate the resulting strength of acid in Normality.**

A. 1 N
B. 0.1 N
C. 0.4 N
D. 0.7 N

86 **Maximum number of nitrogen atom is present in _____**

A. 2 moles NO_2
B. 44.8L of N_2
C. 17 gm of NH_3
D. 6.022×10^{23} molecules of NO_2

87 **The graph representation of P and V at constant temperature is _____**

A. Isobaric
B. Isochoric
C. Adiabatic
D. Isothermal

88 **On heating, saturated solution of NaCl becomes _____**

A. Saturated
B. Unsaturated
C. Super-saturated
D. Remains same

89 **H_2S is gas but H_2O is a liquid, due to _____**

A. Higher molecular weight of H_2O
B. H_2O is more acidic
C. H-bond
D. Electronegativity

90 **Oxidation is _____**

A. Increase of positive valency
B. Decrease of positive valency
C. Gain of electron
D. Decrease in positive charge

91 **Conjugate acid of HCO_3^- is _____**

A. H_2CO_3
B. CO_3^{2-}
C. HCO_2^{-2}
D. HCO_3

92 Conjugate base of HPO_4^{-2} is _____

A. $H_2PO_4^-$ B. H_3PO_4
C. PO_4^{-3} D. H_3O^+

93 Lime water on mixing with CO_2 give milky solution while on excess give _____

A. Soluble bicarbonate
B. Insoluble bicarbonate
C. Soluble carbonate
D. Insoluble carbonate

94 Solubility of carbonates decreases down the magnesium family due to decrease in _____

A. Hydration energy of cations
B. Lattice energy of solids
C. Interionic attractions
D. Entropy of solution formation

95 During an adiabatic process, the volume of a gas is found to be inversely proportional to the square of its absolute temperature. The adiabatic ratio for the gas is _____

A. $\frac{1}{2}$ B. $\frac{3}{2}$
C. $\frac{5}{3}$ D. $\frac{4}{2}$

96 Isotopes of an element differ in _____

A. proton B. nucleus
C. neutron D. positron

97 Building block of protein _____

A. Fatty acid B. Amino acid
C. Lactic acid D. Acetic acid

98 In manufacture of iron, limestone is used as _____

A. Leaching agent B. Oxidizing agent
C. Reducing agent D. Flux

99 At the same temperature and pressure, which of the following gas will have highest KE per mole?

A. Hydrogen B. Nitrogen
C. Helium D. Same for all

100 The equivalent conductance of Ba^{2+} and Cl^- are respectively 127 and 76 $\frac{ohm^{-1} \cdot cm^2}{eq}$ at infinite dilution. What will be the equivalent conductance of $BaCl_2$ at infinite dilution?

A. 203 B. 279
C. 101.5 D. 139.5

101 Common in prokaryotes and eukaryotes _____

A. Nucleus B. Nucleolus
C. Ribosome D. Mitochondria

102 Number of gametes that can be obtained from AaBbCc _____

A. 4 B. 6
C. 8 D. 12

103 A man whose father was colour blind and mother is normal marries a woman who had a colour blind mother and normal father. What percentage of male children of this couple will be colour blind?

A. 25% B. 50%
C. 75% D. 100%

104 Endosperm by fusion of male gamete (AA) and female gamete (aa) in an Angiospermic plant _____

A. Aaa B. AAa
C. AAA D. aaa

105 Which of the following doesn't cause variation?

A. Mutation B. Crossing over
C. Linkage D. Independent assortment

106 Which pigment is dominant in red algae?

A. Chl. a B. Chl. b
C. Phycoerythrin D. Fucoxanthin

107 Fern prothallus is _____

A. Haploid B. Diploid
C. Haplodiploid D. Polyploid

108 Zygospore of spirogyra is _____

A. Haploid B. Diploid
C. Haplo-diploid D. Diplo-diploid

109 Part of plant used in tissue culture is _____

A. In plant B. Out plant
C. Callus D. Explant

110 Hormone that cause rooting _____

A. Auxin B. Cytokinin
C. Gibberellin D. ABA

111 Lichen are pioneer community of _____

A. Hydrosere B. Lithosere
C. Psammosere D. Xerosere

112 Conjoint bicollateral vascular bundle is present in _____

A. Malvaceae B. Leguminosae
C. Compositae D. Cucurbitaceae

113 Transpiration is increased due to _____

A. Light intensity
B. Low temperature
C. High humidity
D. Decrease in wind velocity

114 pH of acid rain _____

A. 5.5-7 B. 3-4.5
C. 7 D. 8

115 In glycolysis, a glucose molecule produces _____

A. 2 pyruvic acid B. Acetyl CoA
C. 4 ATP D. CO_2

116 Turpentine oil is obtained from _____

A. Cycas B. Pinus
C. Ficus D. Fungi

117 All carbohydrates are _____

A. Pentose and hexose
B. Aldose and ketose
C. Deoxyribose and ribose
D. Amine and carboxy group

118 Which of the following is false for Solanaceae?

A. Persistent sepal B. Epipetalous stamen
C. Swollen placenta D. Unilocular ovary

119 Sporophyte of Marchantia is _____

A. Parasitic B. Saprophytic
C. Free living D. Autotrophic

120 Process of conversion of atmospheric N_2 to form usable by plants is called _____

A. Ammonification B. Nitrification
C. Denitrification D. Nitrogen fixation

121 Transgenic plant developed to prevent Vit. A deficiency is _____

A. Corn B. Tomato
C. Potato D. Golden rice

122 Histone protein is _____

A. Acidic B. Basic
C. Neutral D. Amphoteric

123 End product of oxidative phosphorylation _____

A. ATP and H_2O B. ATP
C. ATP, $NADH_2$ and H_2O D. $NADH_2$

124 Cork, cork cambium and secondary cortex are collectively called as _____

A. Phellem B. Phellogen
C. Phelloderm D. Periderm

125 Innermost layer of microspore mother cell _____

A. Epidermis B. Integument
C. Tapetum D. Endothecium

126 Codon for rRNA is CUA. The anti-codon for tRNA is _____

A. CUA B. AUC
C. GAU D. GAT

127 Invert sugar contains _____

A. Glucose and sucrose B. Glucose and fructose
C. Glucose only D. Sucrose only

128 Genetic drift _____

A. Occurs in small population
B. Occurs in large population
C. Occurs in newly formed population
D. Occurs in old population

129 Herkogamy is _____

A. Male and female matures at different time
B. Male mature before female
C. Natural barrier developed between anther and stigma to prevent self pollination
D. Female matures before male

130 If the percentage of A is 30%, what would be the percentage of C?

A. 20% B. 70%
C. 50% D. 25%

131 During G1 phase, amount of DNA = 1.28×10^{-12} gm then amount of DNA at the end of G2 phase is _____

A. 1.28×10^{-12} gm B. 0.64×10^{-12} gm
C. 2.56×10^{-12} gm D. 1.28×10^{-6} gm

132 Which of the following is wrong?

A. Oxidative decarboxylation = mitochondria
B. Photosynthesis = chloroplast
C. Golgi body = protein synthesis
D. Lysosome = Suicidal bag

133 Number of generation needed to divide mitotically to form 32 cells is _____

A. 3 B. 4
C. 6 D. 5

134 Crossing over takes place in _____

A. Sister chromatids of homologous chromosome
B. Non-sister chromatids of homologous chromosomes
C. Sister chromatids of non-homologous chromosomes
D. Non-sister chromatids of non-homologous chromosomes

135 Absence of a chromosome from diploid organism _____

A. Monosomy B. Disomy
C. Trisomy D. Nullisomy

136 **Individual having same alleles for a gene is known as _____**

A. Homozygous
B. Heterozygous
C. Hemizygous
D. Azygous

137 **Glucose units in amylose of starch are joined by _____**

A. Glycosidic bond
B. Peptide bond
C. Ester bond
D. Phosphodiester bond

138 **Area in which a species can live its best life and reproduce is _____**

A. Niche
B. Population
C. Community
D. Habitat

139 **Which term best describes the transfer of energy and nutrients through an ecosystem's interconnected sequences of organisms?**

A. Food web
B. Trophic position
C. Biotic interaction
D. Food chain

140 **Maximum amount of water moves from root to tip of plant due to _____**

A. Transpiration pull
B. Capillarity
C. Root pressure
D. Imbibition

141 **In high altitude, erythrocytes _____**

A. Increase in size
B. Increase in number
C. Decrease in size
D. Decrease in number

142 **A zoology professor found an invertebrate animal which was triploblastic, bilaterally symmetrical and coelomate. What additional feature must be shown by the organism to distinguish it as an Arthropod?**

A. Presence of wings
B. Presence of antenna
C. Presence of shell
D. Presence of jointed appendages

143 **A man used to feel itch in his anal region specially during the night. He might be infected by _____**

A. Tapeworm
B. Hookworm
C. Roundworm
D. Pinworm

144 **Which vitamin helps in maturation of erythrocytes?**

A. Vit. A
B. Vit. B12
C. Vit. C
D. Vit. D

145 **Which of the following is a sexually transmitted disease?**

A. Hepatitis A
B. Hepatitis B
C. Hepatitis C
D. Hepatitis D

146 **Epithelium forming the inner layer of urinary bladder is _____**

A. Simple squamous
B. Simple Columnar
C. Transitional
D. Neurosensory

147 **Pseudostratified epithelium is present in _____**

A. Alveoli
B. Trachea
C. Urinary bladder
D. Pharynx

148 **In earthworm, setae are absent in _____**

A. First segment
B. Last segment
C. Clitellum
D. First segment, last segment and clitellum

149 **Water vascular system is present in _____**

A. Coelenterata
B. Porifera
C. Mollusca
D. Echinodermata

150 **Animals in a higher invertebrate phylum contain a rasping organ in their mouth, mainly associated with processing of food. Identify the organ.**

A. Radula
B. Medulla
C. Gizzard
D. Ctenidia

151 **First enterocoelomic phylum is _____**

A. Aschelminthes
B. Mollusca
C. Chordata
D. Echinodermata

152 **Which of the following cranial nerves has the highest branching?**

A. Facial
B. Abducens
C. Vagus
D. Glossopharyngeal

153 **Among the following, which had a major role in human evolution?**

A. Agriculture
B. Proper use of fire
C. Cave painting
D. Living in group

154 **Major drawback of Darwinism _____**

A. Survival of fittest
B. Variation
C. Struggle for existence
D. Natural selection

155 **Similarity between human and earthworm is, being _____**

A. Ureotelic
B. Uricotelic
C. Ammonotelic
D. Nephrotelic

156 **Vitamin C acts as _____**

A. Agent for integrity of epithelium
B. Blood clotting agent
C. Antioxidant
D. Enzyme for RBC production

157 **Set of viral diseases is _____**

A. Mumps and Rabies
B. Typhoid and Tetanus
C. TB and Cholera
D. Rabies and Syphilis

158 **Response to stimulus of touch is called _____**

A. Rheotaxis
B. Thigmotaxis
C. Thermotaxis
D. Phototaxis

159 **An infant may be feeding entirely on mother's milk which is white in colour but the stool which the infant passes is yellow due to _____**

A. Pancreatic juice poured into duodenum
B. Undigested milk protein
C. Bile pigment passed through bile juice
D. Intestinal juice

160 **Hemozoin is a toxic substance. It is formed after _____**

A. Consumption of whole RBCs
B. Rupture of erythrocytes or destroyed WBCs
C. Consumption of globulin pigment of RBC
D. Consumption of colorful pigment of RBC

161 **In life cycle of Plasmodium, stage with nucleus toward periphery and vacuole in centre is _____**

A. Rosette stage
B. Trophozoite stage
C. Cryptozoite
D. Signet ring

162 **One of the functions of progesterone is _____**

A. Increased action and growth of thyroid
B. Reabsorption of water from collecting duct
C. Stimulation of mammary gland secretion
D. Uterus contraction

163 **Basis of classification in protozoa is _____**

A. Locomotory organelle
B. Size of body
C. Shape
D. Number of nuclei

164 **Both B and T lymphocytes are formed in bone marrow but only T lymphocyte travels to _____**

A. Spleen
B. Thymus
C. Bone marrow
D. Thyroid

165 **Connective tissue in which fat is stored and helps in insulation is _____**

A. Bone
B. Cartilage
C. Adipose tissue
D. Blood

166 **Which of the following is not a bony fish?**

A. Telostomi
B. Osteichthyes
C. Elasmobranchii
D. Lung fish

167 **Type of gland in which whole cell is shedded off along with secretory product _____**

A. Holocrine
B. Apocrine
C. Merocrine
D. Heterocrine

168 **Which of the following flows in the blood vessel of earthworm?**

A. Blood
B. Lymph
C. Hemolymph
D. Serum

169 **Person with sickle cell anaemia is _____**

A. Prone to malaria
B. Less prone to malaria
C. Prone to typhoid
D. Less prone to typhoid

170 **The centrum of 2nd to 7th vertebrae of frog is _____**

A. Anteriorly concave, posteriorly convex
B. Anteriorly concave, posteriorly concave
C. Anteriorly convex, posteriorly concave
D. Anteriorly convex, posteriorly convex

171 **All of the following are characters of chordate except _____**

A. Pharyngeal gill slits
B. Nerve cord
C. Notochord
D. Vertebral column

172 **Which of the following is correctly matched?**

A. Amphetamine: stimulant
B. Heroin: psychotropic
C. LSD: narcotic
D. Benzodiazepines: Pain killer

173 **Which of the following affect joints?**

A. Diabetes mellitus
B. Hypertension
C. Rheumatoid arthritis
D. Osteoporosis

174 **Male genital aperture of earthworm is present in _____**

A. 16th segment
B. 17th segment
C. 18th segment
D. 19th segment

175 **Primary function of smooth muscle is _____**

A. Voluntary action
B. Involuntary action in hollow organs
C. Control the ingestion process
D. Locomotion and voluntary movement of the body parts

176 **A woman shows bilaterally blocked fallopian tubes. Which procedure would you further suggest for her?**

A. Amniocentesis
B. ELISA
C. Antiviral treatment
D. IVF

177 **Why is a frog's mouth difficult to open?**

A. Absence of teeth
B. Due to presence of Prelingual elevation on lower jaw and fossae on upper jaw
C. Upper maxillary teeth
D. Presence of fossa on lower jaw

178 **Transport protein is _____**

A. Albumin
B. Myosin
C. Actin
D. Ricin

179 **A person experiences numbness and swelling in his right arm with frequent tingling sensations in his fingers. He undergoes neurosensory examination at the hospital and was diagnosed with a disorder in skin receptors. Which of the following is correctly matched pair?**

A. Pacinian corpuscle- rapid vibration
B. Merkel cell- slow vibration
C. Meissner corpuscles- stretch
D. Ruffini corpuscles- pain

180 **Homo erectus differ from other hominoids in discovery of _____**

A. Language and communication
B. Use of fire
C. Culture
D. Living in group

181 **How many such pairs of letters in the word "CLEARING" each of which has as many letters between them in the word as in the alphabet?**

A. 0
B. One
C. Two
D. Three

182 **Jenish, Abhaya, Nijan, Madhav and Abhishek are sitting in a circle. Jenish is sitting between Abhaya and Nijan. Madhav is sitting on the right of Abhaya. If position of Madhav is fixed and remaining is rotated in clockwise direction, who's on the left of Madhav?**

A. Jenish
B. Abhaya
C. Nijan
D. Abhishek

183 **Sandesh takes 8 hours to copy a 50-page manuscript while Bipesh can copy the same manuscript in 6 hours. How many hours would it take to copy a similar 100-page manuscript if they work together?**

A. $5\frac{5}{7}$
B. $6\frac{6}{7}$
C. $7\frac{7}{8}$
D. 6

184 **100 cats can kill 100 mice in 100 days. How many days does it take for 4 cats to kill 4 mice?**

A. 4
B. 25
C. 75
D. 100

185 **11 (170) 16, 11 (203) 19, 17 (?) 18**

A. 252
B. 300
C. 306
D. 336

186 **Before 10 years, Sambida's age was four times the age of her daughter. After 10 years Sambida's age will be double the age of her daughter. What is the present age of her daughter?**

A. 20
B. 22
C. 24
D. 26

187 **The average age of 10 students of a class is 18 years. If one more student join them, now the average becomes 19 years. What is the age of the new student?**

A. 21
B. 25
C. 29
D. 31

188 **Two trains having length of 100m and 80m are moving in same direction, with speeds $51\ \frac{m}{s}$ and $42\ \frac{m}{s}$ respectively. In what time will they cross each other?**

A. 18 seconds
B. 20 seconds
C. 25 seconds
D. 30 seconds

189 **When 3 cubes of dimension 5 cm are melted to form a wire of diameter 2 cm, what would be the length of the wire?**

A. 59.6 cm
B. 119.3 cm
C. 178.8 cm
D. 238.6 cm

190 **In a library, the number of Science, Mathematics and Computer books are in the ratio of 5:6:2. The number of English books is two-third the number of Computer books. If the total number of books is 473, then what is the number of English books in the library?**

A. 11
B. 44
C. 66
D. 165

191 **Five students Ramesh, Santosh, Sanjeev, Sumit and Naresh are sitting in a line. Ramesh is not adjacent to Naresh or Sanjeev. Ramesh is adjacent to Sumit. Sumit is at the middle and adjacent to Sanjeev in the row. Then, who is between Naresh and Sumit?**

A. Sumit
B. Ramesh
C. Sanjeev
D. Naresh

192 **Find the missing value.**

I	M
2	3
J	R
2	9
T	U
4	?

A. 1
B. 2
C. 3
D. 4

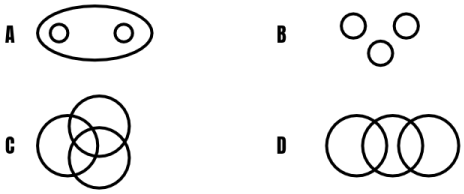
193 **LIGHT:L9I6G5H8T0::WHIT:?**

A. W7H4I6T2
B. W3H8I6T0
C. W5H4I9T6
D. W1H8I9T4

194 **Mirror image of "dIffi3uLtTi5e$" is _____**

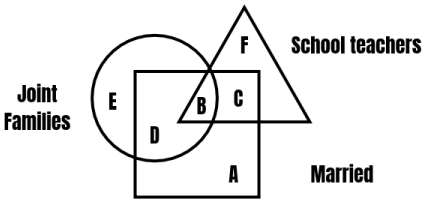
A. ƨɘ5iTʇLuƐiffIb
B. ɘlʇ|:IƆ0Ր†|I262
C. ƨɘƨiTʇLuƐiffIb
D. dIffi3uLtTi5e$

195 **Which of the following diagrams indicates the best relation between Teacher, Men and Women?**



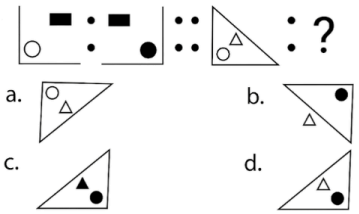
A.
B.
C.
D.

196 **Triangle represents the school teachers. Square represents married persons. Circle represents persons living in joint families. Married persons living in joint families but not working as the school teachers are represented by _____**



A. D
B. F
C. A
D. C

197 **Select the figure that will replace the question mark (?):**

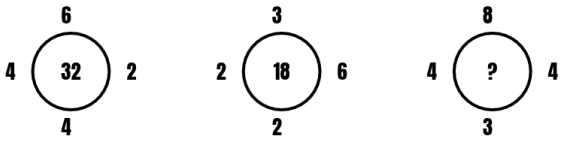


A.
B.
C.
D.

198 **1, 22, 2, 20, 4, 18, 8, 16, 16, 14, ..., ...**

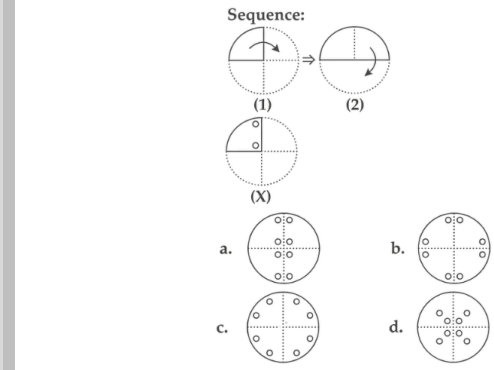
A. 32, 12
B. 32, 10
C. 24, 8
D. 16, 10

199 **Find the missing number in the given figure.**



A. 19
B. 40
C. 46
D. 56

200 **Find out how the pattern would appear when the following folded paper 'X' is opened in the given sequence.**



A.
B.
C.
D.